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**Uncertainty-Aware Model Predictive  
Control for Constrained Nonlinear  
Systems: A UAV Landing Case Study**

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# Uncertainty-Aware Model Predictive Control for Constrained Nonlinear Systems: A UAV Landing Case Study

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## Abstract

Autonomous unmanned aerial vehicles (UAVs) are increasingly deployed in safety-critical applications such as precision landing on dynamic platforms, where robust control strategies are essential for ensuring safe and reliable operation. This research proposes an uncertainty-aware Model Predictive Control (MPC) framework tailored for constrained nonlinear UAV landing, addressing the challenge of handling environmental disturbances and model uncertainties during the landing process. The proposed framework extends traditional MPC by incorporating robust constraint tightening to explicitly account for bounded uncertainties, thereby ensuring that state and input constraints are respected even under disturbance conditions. The research develops the uncertainty-aware MPC formulation, which integrates structured uncertainty modeling, robust constraint enforcement, and real-time feasibility considerations. Through simulation-based evaluation, the framework is compared against baseline MPC controllers in nominal and uncertain landing scenarios. Key performance metrics, such as landing position error, touchdown velocity, constraint violations, and computational feasibility, are analyzed. The results demonstrate that the proposed controller is capable of achieving safe landing conditions under uncertainty while maintaining constraint satisfaction and real-time computational performance. Sensitivity analysis further highlights the trade-offs between robustness and performance, showing that the proposed framework offers a reliable solution for UAV landing tasks in dynamic environments. This work lays the groundwork for future research into adaptive uncertainty handling and real-world implementation, aiming to enhance the safety and efficiency of autonomous UAV systems in uncertain operating conditions.

# Contents

|                                                               |           |
|---------------------------------------------------------------|-----------|
| <b>List of Tables</b>                                         | <b>4</b>  |
| <b>List of Figures</b>                                        | <b>4</b>  |
| <b>1 Motivation and Background</b>                            | <b>5</b>  |
| 1.1 Motivation . . . . .                                      | 5         |
| 1.2 Background . . . . .                                      | 5         |
| 1.2.1 Problem Statement . . . . .                             | 6         |
| 1.3 Research Objectives . . . . .                             | 6         |
| <b>2 Overview and Scope</b>                                   | <b>7</b>  |
| 2.1 Critical Synthesis and Research Gaps . . . . .            | 7         |
| <b>3 Overview of the Methodological Approach</b>              | <b>8</b>  |
| 3.1 Structured Uncertainty Modeling . . . . .                 | 9         |
| 3.2 Uncertainty-Aware MPC Formulation . . . . .               | 9         |
| 3.2.1 Nominal Nonlinear MPC Problem . . . . .                 | 10        |
| 3.2.2 Robust Constraint Handling via Tube-Based MPC . . . . . | 10        |
| 3.2.3 Treatment of Stochastic Uncertainty . . . . .           | 10        |
| 3.2.4 Terminal Conditions and Recursive Feasibility . . . . . | 10        |
| 3.2.5 Optional Safety Augmentation . . . . .                  | 11        |
| <b>4 System Model and Assumptions</b>                         | <b>11</b> |
| 4.1 Modeling . . . . .                                        | 11        |
| 4.1.1 UAV Dynamic Model . . . . .                             | 11        |
| 4.1.2 Operating Regime and Scope . . . . .                    | 12        |
| 4.1.3 Assumptions . . . . .                                   | 12        |
| 4.2 Nominal Model Predictive Control Formulation . . . . .    | 12        |
| 4.2.1 Prediction Model . . . . .                              | 13        |
| 4.2.2 Cost Function . . . . .                                 | 13        |
| 4.2.3 Constraints . . . . .                                   | 13        |
| 4.2.4 Nominal MPC Optimization Problem . . . . .              | 13        |
| 4.3 Uncertainty Modeling and Error Dynamics . . . . .         | 14        |
| 4.3.1 Uncertain System Model . . . . .                        | 14        |
| 4.3.2 Error Dynamics under Feedback Control . . . . .         | 14        |
| 4.3.3 Assumptions on Error Boundedness . . . . .              | 15        |
| 4.4 Robust Constraint Tightening Strategy . . . . .           | 15        |
| 4.4.1 Motivation for Constraint Tightening . . . . .          | 15        |
| 4.4.2 Tightened State and Input Constraints . . . . .         | 15        |
| 4.4.3 Integration into the MPC Optimization Problem . . . . . | 15        |
| 4.4.4 Conservatism and Design Trade-offs . . . . .            | 16        |
| 4.5 Terminal Conditions and Feasibility Discussion . . . . .  | 16        |
| 4.5.1 Role of Terminal Conditions in UAV Landing . . . . .    | 16        |
| 4.5.2 Terminal Cost and Terminal Constraint Set . . . . .     | 16        |
| 4.5.3 Recursive Feasibility Considerations . . . . .          | 17        |
| 4.5.4 Scope and Limitations . . . . .                         | 17        |
| 4.6 Implementation Considerations . . . . .                   | 17        |
| 4.6.1 Optimization Problem Structure . . . . .                | 18        |
| 4.6.2 Real-Time Feasibility Considerations . . . . .          | 18        |
| 4.6.3 Numerical Implementation Scope . . . . .                | 18        |
| 4.6.4 Transition to Simulation and Evaluation . . . . .       | 19        |

|          |                                                  |           |
|----------|--------------------------------------------------|-----------|
| <b>5</b> | <b>Simulation Results and Discussion</b>         | <b>19</b> |
| 5.1      | Simulation Environment and Models . . . . .      | 19        |
| 5.2      | Landing Scenario Description . . . . .           | 20        |
| 5.3      | Baseline Controllers . . . . .                   | 20        |
| 5.4      | Evaluation Metrics . . . . .                     | 21        |
| 5.4.1    | Constraint Satisfaction Metrics . . . . .        | 21        |
| 5.4.2    | Performance Metrics . . . . .                    | 21        |
| 5.4.3    | Robustness Metrics . . . . .                     | 21        |
| 5.4.4    | Computational Feasibility Metrics . . . . .      | 22        |
| 5.5      | Nominal Landing Results . . . . .                | 22        |
| 5.5.1    | Simulation Setup . . . . .                       | 22        |
| 5.5.2    | Performance Results . . . . .                    | 22        |
| 5.6      | Landing Under Uncertainty . . . . .              | 22        |
| 5.6.1    | Simulation Setup . . . . .                       | 23        |
| 5.6.2    | Performance Results . . . . .                    | 23        |
| 5.7      | Robustness and Sensitivity Analysis . . . . .    | 24        |
| 5.7.1    | Sensitivity to Uncertainty Bounds . . . . .      | 24        |
| 5.7.2    | Robustness to Disturbance Realizations . . . . . | 24        |
| 5.8      | Discussion of Trade-offs . . . . .               | 24        |
| 5.9      | Project Timeline . . . . .                       | 25        |
| <b>6</b> | <b>Conclusions</b>                               | <b>25</b> |
| 6.1      | Summary of Contributions . . . . .               | 25        |
| 6.2      | Key Findings . . . . .                           | 26        |
| 6.3      | Limitations and Future Work . . . . .            | 26        |
|          | <b>References</b>                                | <b>27</b> |

## List of Tables

|   |                                                                                         |    |
|---|-----------------------------------------------------------------------------------------|----|
| 1 | Summary of MPC formulations for UAV landing with focus on uncertainty handling. . . . . | 8  |
| 2 | Summary of MPC formulations for UAV landing with focus on uncertainty handling. . . . . | 19 |
| 3 | Controllers implemented for comparative evaluation. . . . .                             | 23 |
| 4 | Proposed research timeline and activities over a 12-month period. . . . .               | 25 |

## List of Figures

|   |                                                                                          |    |
|---|------------------------------------------------------------------------------------------|----|
| 1 | Conceptual architecture of the proposed uncertainty-aware MPC framework for UAV landing. | 11 |
| 2 | Illustrative UAV landing scenario and constraint geometry. . . . .                       | 17 |

# 1 Motivation and Background

## 1.1 Motivation

Unmanned aerial vehicles have become increasingly important in applications that require high levels of autonomy, reliability, and safety, including maritime operations, emergency response, surveillance, logistics, and infrastructure inspection. In many of these applications, autonomous landing represents a mission-critical maneuver where failure can result in vehicle damage, mission loss, or safety hazards to surrounding infrastructure and personnel. As a result, autonomous UAV landing is not only a problem of control performance but also a safety-critical systems challenge.

Among all phases of UAV operation, landing is particularly demanding due to the interaction of strongly nonlinear vehicle dynamics, strict state and actuator constraints, and external disturbances that intensify near the ground or landing surface. These challenges are further exacerbated when the landing surface itself is dynamic, such as ship decks or moving ground platforms, where environmental disturbances and prediction errors directly affect feasibility and safety [1]. Even small deviations during the final descent phase can lead to constraint violations or unsafe touchdown conditions, highlighting the need for control strategies that explicitly account for uncertainty.

Model Predictive Control has emerged as a dominant framework for autonomous UAV landing due to its ability to explicitly handle constraints while optimizing performance objectives over a finite prediction horizon. Recent surveys identify MPC as the most widely adopted control approach for UAV landing across a range of scenarios, including static landings, maritime recovery, and landings on moving platforms [1]. Compared to classical PID or linear controllers, MPC-based approaches offer superior constraint handling and predictive capabilities, particularly for nonlinear UAV dynamics.

However, the majority of MPC-based UAV landing controllers rely on deterministic system models and assume perfect knowledge of dynamics and disturbances. In realistic operating environments, UAV landing is affected by multiple sources of uncertainty, including wind gusts, aerodynamic ground effects, payload-induced parameter variations, and sensor noise. When such uncertainties are neglected or handled through fixed safety margins, deterministic MPC can exhibit degraded performance, excessive conservatism, or loss of safety during landing.

These limitations motivate the development of control strategies that explicitly reason about uncertainty while preserving the constraint handling and predictive advantages of MPC. In particular, there is a need for uncertainty-aware MPC formulations that are specifically tailored to safety-critical UAV landing and capable of balancing robustness, performance, and real-time feasibility under realistic uncertainty conditions.

## 1.2 Background

From a control-theoretic perspective, autonomous UAV landing can be formulated as a constrained nonlinear optimal control problem subject to uncertainty. The system dynamics are nonlinear, the admissible state and input sets are tightly bounded, and the control objective involves tracking a landing trajectory while satisfying terminal touchdown conditions. These characteristics make UAV landing a representative benchmark for advanced predictive control methods.

Model Predictive Control addresses such problems by solving a finite-horizon optimal control problem at each control update and applying the first control input in a receding-horizon fashion. This structure enables MPC to anticipate future constraint violations, incorporate predictions of platform motion or reference trajectories, and adjust control actions proactively. The explicit inclusion of state and input constraints within the optimization problem is a key reason for MPC’s widespread adoption in UAV landing applications [1,9].

Uncertainty-aware extensions of MPC aim to improve robustness by incorporating uncertainty directly into the prediction and optimization process. Robust MPC formulations consider bounded disturbances and model uncertainty and seek to guarantee constraint satisfaction for all admissible uncertainty realizations, often using tube-based or invariant set concepts [2]. These approaches provide strong safety guarantees but typically rely on worst-case assumptions that can introduce conservatism, which is particularly undesirable during precision landing maneuvers [3].

Stochastic MPC formulations represent uncertainty probabilistically and enforce chance constraints that are satisfied with a specified confidence level. This allows explicit trade-offs between performance and risk and can be effective when disturbance statistics are known or can be estimated reliably [4]. However, during

UAV landing, disturbance characteristics may vary rapidly due to aerodynamic effects and environmental interactions, making accurate probabilistic modeling challenging and increasing computational complexity.

More recently, learning-augmented MPC approaches have been proposed to reduce model mismatch by incorporating data-driven models or uncertainty estimates obtained from Gaussian processes or neural networks [5]. While these methods have demonstrated improved adaptability and predictive accuracy, their use in safety-critical landing scenarios raises concerns regarding formal safety guarantees and certification.

Despite significant progress, existing MPC-based UAV landing approaches often address uncertainty sources in isolation or rely on assumptions that limit practical applicability. In particular, there remains a gap between theoretical developments in uncertainty-aware MPC and the requirements of constrained nonlinear UAV landing, where multiple uncertainty sources interact and strict safety constraints must be enforced in real time. Addressing this gap motivates the research problem formulated in the following section.

### 1.2.1 Problem Statement

Model Predictive Control is widely used for autonomous UAV landing because it enables explicit handling of nonlinear dynamics and strict state and input constraints [1]. However, most existing MPC-based landing controllers rely on deterministic models or fixed safety margins that do not adequately capture the uncertainty present in realistic landing environments. In practice, UAV landing is affected by multiple interacting sources of uncertainty, including wind disturbances, aerodynamic modeling errors, parametric variations, and sensor noise, which can lead to degraded performance or constraint violations if not explicitly addressed. Robust MPC formulations provide safety guarantees under bounded uncertainty but often introduce excessive conservatism that limits landing precision and efficiency for nonlinear systems [2, 3]. Stochastic and learning-based MPC approaches offer alternative uncertainty representations but depend on accurate uncertainty modeling and raise challenges related to computational complexity and safety assurance in safety-critical landing scenarios [4, 5]. Consequently, there is a lack of a control framework that is specifically tailored to constrained nonlinear UAV landing and capable of explicitly incorporating uncertainty while balancing robustness, performance, strict constraint satisfaction, and real-time feasibility. Addressing this gap constitutes the central problem investigated in this research.

## 1.3 Research Objectives

The objective of this research is to investigate and validate an uncertainty-aware Model Predictive Control framework for constrained nonlinear systems, using autonomous UAV landing as a representative safety-critical case study. The research focuses on explicit uncertainty handling while maintaining strict constraint satisfaction and practical computational feasibility.

The specific objectives of this research are as follows:

1. **To develop a constrained nonlinear MPC formulation for UAV landing that explicitly incorporates uncertainty.**

This objective focuses on formulating a nonlinear predictive control problem that accounts for bounded disturbances and parametric uncertainty within the system model, rather than relying on purely deterministic assumptions or fixed safety margins.

2. **To design a robustness mechanism based on constraint tightening for uncertainty handling during landing.**

The research aims to implement a tube-based or set-based robustness strategy that enforces state and input constraints under bounded uncertainty, with particular attention to feasibility and landing precision in nonlinear UAV dynamics.

3. **To investigate the impact of uncertainty-aware control on landing performance and safety.**

Through simulation-based evaluation, the research examines how explicit uncertainty handling influences landing accuracy, constraint violations, and control effort when compared to deterministic MPC and classical baseline controllers.

4. **To assess the computational feasibility of the proposed uncertainty-aware MPC framework for real-time landing scenarios.**

This objective evaluates solver performance, prediction horizon selection, and execution time to determine whether the proposed formulation can operate within realistic computational limits for UAV landing.

5. **To analyze the sensitivity of controller performance to uncertainty modeling assumptions.**

The research studies how variations in uncertainty bounds and modeling assumptions affect robustness and performance, providing insight into the practical reliability of uncertainty-aware MPC designs.

Together, these objectives define a focused and achievable research scope that addresses the identified gap between deterministic MPC approaches and the requirements of constrained nonlinear UAV landing under uncertainty.

## 2 Overview and Scope

This literature review examines research on Model Predictive Control for constrained nonlinear systems under uncertainty, with a specific focus on autonomous UAV landing as a safety-critical application. The purpose of the review is not to provide an exhaustive survey of all MPC variants, but to establish the theoretical and practical context necessary to motivate the proposed uncertainty-aware MPC framework and to identify gaps relevant to constrained nonlinear landing problems.

UAV landing presents a representative benchmark for uncertainty-aware control due to the combination of nonlinear dynamics, strict state and actuator constraints, environmental disturbances, and the requirement for precise terminal behavior. Recent surveys indicate that MPC-based approaches dominate the UAV landing literature because of their ability to explicitly encode constraints and anticipate future system behavior [1]. However, these studies also highlight that uncertainty is often treated implicitly or through conservative safety margins, which can limit performance during precision landing maneuvers.

The scope of this review is therefore limited to MPC formulations that explicitly address uncertainty in a manner relevant to nonlinear UAV landing. This includes robust MPC approaches that guarantee constraint satisfaction under bounded uncertainty, stochastic MPC formulations that incorporate probabilistic representations of disturbances, and related extensions that aim to balance robustness and performance [2–4]. Learning-based and safety-integrated MPC methods are discussed only insofar as they inform current limitations in uncertainty handling and safety assurance for landing applications [5, 6].

Rather than reviewing all UAV control strategies, this section focuses on how different uncertainty-aware MPC formulations handle constraints, robustness, and computational complexity in the context of landing. The review emphasizes limitations that motivate the present research, namely the lack of MPC frameworks that are specifically tailored to constrained nonlinear UAV landing and capable of explicitly incorporating uncertainty while maintaining strict constraint enforcement and real-time feasibility. This focused scope ensures that the literature review directly supports the problem formulation and research objectives outlined in Section 1.

### 2.1 Critical Synthesis and Research Gaps

The reviewed literature demonstrates that Model Predictive Control has become the dominant framework for autonomous UAV landing due to its ability to explicitly handle nonlinear dynamics and strict state and input constraints [1]. Significant progress has been made in applying nonlinear MPC to precision landing scenarios, including landings on static and moving platforms, with demonstrated improvements in tracking accuracy and constraint handling compared to classical control methods [6, 9]. However, the treatment of uncertainty remains a key limitation across much of the existing work.

Robust MPC formulations address uncertainty by guaranteeing constraint satisfaction under bounded disturbances and model mismatch, providing strong safety assurances that are desirable for landing maneuvers [2]. Nevertheless, these approaches often rely on worst-case assumptions that introduce conservatism, which can reduce landing precision and increase control effort for nonlinear UAV systems [3]. This conservatism

is particularly problematic during the final descent phase, where tight constraints and accurate touchdown conditions are required.

Stochastic MPC approaches offer an alternative by modeling uncertainty probabilistically and enforcing chance constraints, enabling explicit trade-offs between performance and risk [4]. While attractive in principle, these methods depend on accurate disturbance statistics and incur increased computational complexity, which limits their practical applicability in rapidly changing landing environments. Learning-augmented MPC frameworks further improve predictive accuracy and adaptability by reducing model mismatch through data-driven techniques [5]. However, their reliance on learned models raises unresolved concerns regarding safety guarantees, interpretability, and certification in safety-critical landing applications.

Safety-integrated MPC approaches, such as those incorporating Control Barrier Functions, have demonstrated strong safety performance in landing scenarios by explicitly enforcing invariance of safe sets [6]. Despite their effectiveness, these methods are most often applied under deterministic assumptions and are rarely combined with explicit uncertainty handling. Similarly, distributed and cooperative MPC formulations extend MPC-based landing to more complex operational scenarios but introduce additional sources of uncertainty related to communication and platform coupling [7].

Overall, the literature indicates that existing MPC-based UAV landing approaches typically address uncertainty, safety, and computational feasibility in isolation rather than within a unified framework. In particular, there remains a gap in MPC formulations that are specifically tailored to constrained nonlinear UAV landing and capable of explicitly incorporating uncertainty while maintaining strict constraint enforcement and practical real-time feasibility. Addressing this gap motivates the uncertainty-aware MPC framework proposed in this research, which focuses on structured uncertainty handling and robust constraint satisfaction within a nonlinear predictive control setting.

Table 1: Summary of MPC formulations for UAV landing with focus on uncertainty handling.

| Reference                 | MPC Type          | Uncertainty        | Constraints   | Key Findings                         | Limitations                          |
|---------------------------|-------------------|--------------------|---------------|--------------------------------------|--------------------------------------|
| Panjavarnam et al., 2025  | Survey            | Varies             | Varies        | MPC dominant for constraint handling | Limited unified uncertainty handling |
| Batool et al., 2025       | Nonlinear MPC     | Deterministic      | Hard          | High precision landing               | No explicit uncertainty modeling     |
| Leeman et al., 2025       | Robust NMPC       | Bounded            | Robust        | Guaranteed constraint satisfaction   | Conservative, high computation       |
| Aryan, 2025               | Stochastic NMPC   | Chance-constrained | Probabilistic | Good risk-performance trade-off      | Needs accurate statistics, costly    |
| Wehbeh and Kerrigan, 2025 | Robust MPC        | Sensitivity-based  | Hard          | Reduced conservatism                 | Limited to simple maneuvers          |
| Saviolo et al., 2024      | Learning MPC      | Data-driven        | Mixed         | Improved adaptability                | Limited safety guarantees            |
| Stephenson et al., 2024   | Distributed MPC   | Learned motion     | Coupled       | Cooperative landing                  | Communication challenges             |
| Ru and Subbarao, 2017     | Nonlinear control | Implicit           | Deterministic | Stable control                       | Not for precision landing            |

### 3 Overview of the Methodological Approach

This research adopts a model-based control methodology focused on the design and evaluation of an uncertainty-aware Model Predictive Control framework for constrained nonlinear systems, with autonomous UAV landing serving as the primary case study. The methodological approach is structured to directly address the research problem identified in Section 1, namely the lack of MPC formulations that explicitly account for uncertainty while maintaining strict constraint satisfaction and practical real-time feasibility in nonlinear landing scenarios.

The core of the proposed approach is a constrained nonlinear MPC formulation that incorporates uncertainty through structured modeling and robust constraint handling. Rather than relying on purely deterministic assumptions or overly conservative safety margins, the methodology explicitly represents key sources of uncertainty relevant to UAV landing, including bounded disturbances and parametric variations. These uncertainties are integrated into the predictive control framework using robustness mechanisms that are compatible with nonlinear dynamics and safety-critical constraints.

To ensure safety during landing, the methodology emphasizes robust constraint enforcement using set-based techniques, such as constraint tightening derived from bounded uncertainty representations. This allows state and input constraints to be respected despite disturbances and modeling errors, which is essential during the final descent and touchdown phases. Probabilistic uncertainty representations are considered at an analytical and evaluative level to assess robustness and performance sensitivity, rather than forming the primary basis of the control law, thereby maintaining methodological focus and computational tractability.

The UAV landing task is used throughout the methodology to ground theoretical developments in a realistic application. The nonlinear system model, constraints, uncertainty representations, and performance metrics are all defined with reference to the physical and operational requirements of autonomous landing. High-fidelity simulation studies are employed to evaluate controller behavior under increasing uncertainty and to compare the proposed uncertainty-aware MPC framework against deterministic MPC, robust MPC, and classical baseline controllers.

Overall, the methodological approach is designed to balance rigor and feasibility. It prioritizes explicit uncertainty handling and safety-oriented constraint enforcement while remaining achievable within the scope of a Master’s research project. The structure of the methodology ensures that theoretical design choices are consistently validated through simulation-based analysis, providing insight into the practical benefits and limitations of uncertainty-aware MPC for constrained nonlinear UAV landing.

### 3.1 Structured Uncertainty Modeling

Uncertainty plays a central role in autonomous UAV landing due to environmental disturbances, modeling inaccuracies, and imperfect state information. In this research, uncertainty is modeled in a structured and application-driven manner to support robust constraint enforcement within a nonlinear MPC framework, while avoiding unnecessary complexity that would compromise feasibility.

The uncertainty affecting the UAV landing task is classified into three main categories based on its physical origin and impact on control performance. The first category is *parametric uncertainty*, which captures variations in system parameters such as mass, inertia, and aerodynamic coefficients arising from payload changes, battery depletion, or modeling simplifications. These parameters are assumed to lie within known bounded sets that reflect realistic operating conditions.

The second category consists of *additive disturbances* acting on the system dynamics. These disturbances represent environmental effects such as wind gusts, turbulence, and near-ground aerodynamic interactions. For control design purposes, these disturbances are modeled as bounded signals with known limits, allowing their effects to be explicitly accounted for through robust constraint handling techniques. This bounded disturbance representation forms the primary uncertainty model used within the MPC formulation to ensure safety and constraint satisfaction during landing.

The third category is *state estimation uncertainty* resulting from sensor noise and imperfect state reconstruction. Rather than modeling estimation uncertainty explicitly within the MPC optimization problem, its effects are captured implicitly through conservative bounds on the system state or disturbance terms, consistent with standard robust MPC assumptions.

While some uncertainty sources, such as wind turbulence and sensor noise, may exhibit stochastic characteristics, probabilistic representations are not used as the primary basis for constraint enforcement in this work. Instead, stochastic uncertainty models are employed at an analytical and evaluative level to assess controller robustness and performance sensitivity under randomized uncertainty realizations. This design choice prioritizes safety and tractability by ensuring that hard state and input constraints are enforced under bounded uncertainty assumptions.

### 3.2 Uncertainty-Aware MPC Formulation

This section presents the formulation of the uncertainty-aware Model Predictive Control framework proposed in this research. The formulation is centered on a constrained nonlinear MPC design with explicit robustness mechanisms to handle bounded uncertainty, ensuring strict enforcement of state and input constraints during UAV landing. The approach prioritizes safety and feasibility under uncertainty while maintaining computational tractability suitable for real-time implementation.

### 3.2.1 Nominal Nonlinear MPC Problem

The starting point of the formulation is a nominal nonlinear MPC problem defined for the uncertainty-free system dynamics. At each control update, an optimal control sequence is computed by minimizing a finite-horizon cost function of the form

$$\min_{u_0, \dots, u_{N-1}} \sum_{i=0}^{N-1} \ell(x_i, u_i) + \ell_f(x_N)$$

subject to the nominal discrete-time dynamics

$$x_{i+1} = f_d(x_i, u_i),$$

and the state and input constraints

$$x_i \in \mathcal{X}, \quad u_i \in \mathcal{U}.$$

The stage cost penalizes tracking error and control effort, while the terminal cost encourages convergence toward desired landing conditions. This nominal MPC problem provides the baseline structure upon which uncertainty handling mechanisms are incorporated.

### 3.2.2 Robust Constraint Handling via Tube-Based MPC

To ensure constraint satisfaction under bounded uncertainty, the nominal MPC formulation is augmented using a tube-based robust MPC approach. The actual system state is decomposed into a nominal component and an error component induced by disturbances and modeling uncertainty. A stabilizing feedback law is used to bound the evolution of this error within a robust invariant set.

Constraint tightening is applied to the nominal state and input trajectories such that, if the nominal predictions satisfy the tightened constraints, the true system trajectory satisfies the original constraints for all admissible uncertainty realizations. This mechanism enables strict enforcement of safety-critical constraints during landing despite bounded disturbances and parameter variations.

For the nonlinear UAV dynamics considered in this work, invariant error sets are approximated using local linearization around the landing trajectory and offline computation of stabilizing feedback gains. This design choice balances robustness and computational feasibility and is consistent with established nonlinear robust MPC practices.

### 3.2.3 Treatment of Stochastic Uncertainty

Certain uncertainty sources relevant to UAV landing, such as wind turbulence and sensor noise, may exhibit stochastic characteristics. In this research, such stochastic uncertainty is not incorporated directly into the MPC constraints or optimization problem. Instead, it is used to evaluate the robustness and performance sensitivity of the proposed controller through randomized simulation studies.

This separation ensures that hard state and input constraints are enforced under bounded uncertainty assumptions while still allowing insight into controller behavior under probabilistic disturbance realizations. By avoiding direct integration of chance constraints into the core MPC formulation, the approach maintains theoretical clarity and avoids unnecessary computational burden.

### 3.2.4 Terminal Conditions and Recursive Feasibility

To promote stability and recursive feasibility, terminal conditions are incorporated into the MPC formulation. A terminal constraint set is defined such that, once the system state enters this set, there exists an admissible control action that keeps the state within the set under bounded uncertainty. The terminal cost and terminal constraint are designed consistently with the robust constraint tightening strategy to ensure feasibility over successive MPC iterations.

These terminal conditions are particularly important for UAV landing, where convergence to safe touch-down states must be guaranteed within a finite horizon.

### 3.2.5 Optional Safety Augmentation

For landing scenarios involving obstacles or complex environment constraints, additional safety requirements may be incorporated into the MPC formulation through auxiliary state constraints. These safety mechanisms are treated as optional extensions and are only considered if they can be integrated without compromising feasibility or real-time performance. They do not form a core contribution of this research and are included to demonstrate extensibility of the proposed framework.

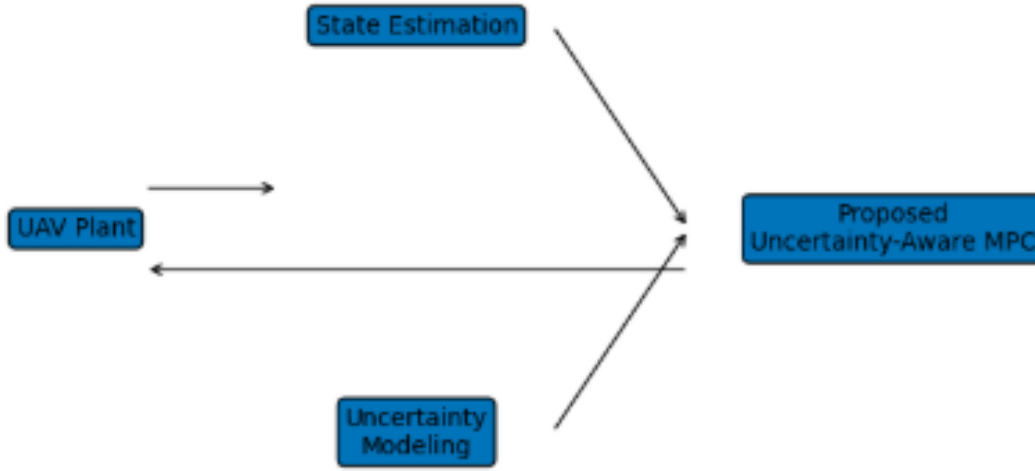


Figure 1: Conceptual architecture of the proposed uncertainty-aware MPC framework for UAV landing.

As shown in Figure 1, the proposed framework integrates state estimation and structured uncertainty modeling into a nonlinear Model Predictive Control scheme. This conceptual architecture highlights how uncertainty information is intended to inform the predictive optimization process while maintaining feedback from the UAV system. The figure is illustrative and is not intended to represent a finalized implementation.

## 4 System Model and Assumptions

### 4.1 Modeling

This chapter presents the control design and analysis of the proposed uncertainty-aware Model Predictive Control framework for autonomous UAV landing. The focus is on translating the methodological concepts introduced in Chapter 3 into a concrete control formulation suitable for robustness analysis and simulation-based evaluation. This section introduces the system model used for control design and clearly states the assumptions under which the proposed approach is developed.

#### 4.1.1 UAV Dynamic Model

The UAV considered in this research is modeled as a rigid-body system with nonlinear translational and rotational dynamics. The model captures the essential dynamics relevant to the landing phase, including position, velocity, attitude, and angular velocity states. Such nonlinear models are widely adopted in MPC-based UAV landing studies due to their ability to represent coupling effects and constraint-relevant behaviors during descent and touchdown [1, 6].

Let the system state vector be defined as

$$x = \begin{bmatrix} p^\top & v^\top & \eta^\top & \omega^\top \end{bmatrix}^\top,$$

where  $p \in \mathbb{R}^3$  denotes the position of the UAV in an inertial frame,  $v \in \mathbb{R}^3$  represents translational velocity,  $\eta \in \mathbb{R}^3$  denotes attitude variables, and  $\omega \in \mathbb{R}^3$  represents angular velocity. The control input vector consists of thrust and attitude-related inputs subject to actuator limitations.

The continuous-time nonlinear dynamics are expressed in compact form as

$$\dot{x}(t) = f(x(t), u(t)) + w(t),$$

where  $u(t)$  is the control input and  $w(t)$  represents additive disturbances capturing modeling errors and environmental effects. For use within the MPC framework, the system dynamics are discretized using a fixed sampling time, yielding a discrete-time nonlinear model suitable for prediction over a finite horizon [2].

This model is intended to capture the dominant dynamics relevant to autonomous landing while remaining computationally tractable for real-time predictive control.

#### 4.1.2 Operating Regime and Scope

The control design focuses exclusively on the landing phase of UAV operation. This includes controlled descent, approach, and touchdown within a constrained operating envelope. Aggressive maneuvers, high-speed flight, and fault conditions are outside the scope of this work. The UAV is assumed to operate near hover or low-velocity descent, which is consistent with standard assumptions in MPC-based landing studies [1, 9].

By restricting the operating regime, the proposed control framework can exploit local properties of the nonlinear dynamics to balance robustness and performance without introducing unnecessary conservatism.

#### 4.1.3 Assumptions

The following assumptions are made throughout the control design and analysis:

1. **State Availability:** All system states required by the MPC controller are assumed to be available through onboard sensors and state estimation algorithms. Estimation errors are treated implicitly as bounded uncertainty, consistent with robust MPC assumptions [2].
2. **Bounded Uncertainty:** External disturbances and model uncertainties affecting the UAV dynamics are assumed to be bounded within known limits. These bounds are selected to reflect realistic landing conditions, including wind disturbances and modeling inaccuracies.
3. **Known Constraints:** State and input constraints, including actuator limits and safety-related state bounds, are assumed to be known a priori and remain constant during the landing maneuver.
4. **Fixed Sampling Time:** The MPC controller operates with a fixed sampling time that is sufficiently small to capture the relevant dynamics of the landing phase and to support real-time optimization.

These assumptions are standard in robust nonlinear MPC formulations and are necessary to ensure feasibility and safety during constrained UAV landing [2, 3].

## 4.2 Nominal Model Predictive Control Formulation

This section presents the nominal Model Predictive Control formulation that serves as the baseline for the uncertainty-aware control design developed in the subsequent sections. The nominal MPC problem is defined for the nonlinear UAV dynamics introduced in Section 4.1 and does not yet account for uncertainty or robustness. This separation allows the effects of uncertainty handling to be introduced systematically while maintaining conceptual clarity.

#### 4.2.1 Prediction Model

The nominal discrete-time system dynamics used for prediction within the MPC framework are given by

$$x_{k+1} = f_d(x_k, u_k),$$

where  $x_k$  denotes the system state at time step  $k$ ,  $u_k$  is the control input, and  $f_d(\cdot)$  represents the discretized nonlinear UAV dynamics. The discretization is performed using a fixed sampling time consistent with the assumptions stated in Section 4.1. This nonlinear prediction model captures the dominant dynamics relevant to the landing phase and is commonly used in MPC-based UAV landing control [1, 6].

The prediction model is assumed to be known and deterministic in this nominal formulation. Modeling errors and disturbances are not considered at this stage and will be addressed explicitly in Sections 4.3 and 4.4.

#### 4.2.2 Cost Function

At each control update, the nominal MPC computes a control input sequence by minimizing a finite-horizon cost function of the form

$$J = \sum_{i=0}^{N-1} \ell(x_i, u_i) + \ell_f(x_N),$$

where  $N$  is the prediction horizon,  $\ell(\cdot)$  is the stage cost, and  $\ell_f(\cdot)$  is the terminal cost. The stage cost penalizes deviations from the desired landing trajectory as well as control effort, while the terminal cost promotes convergence toward a target landing state.

The structure of the cost function follows standard MPC practice and reflects typical performance objectives for UAV landing, including tracking accuracy, smooth control action, and stable touchdown behavior [1, 9]. The specific weighting matrices are selected based on the landing requirements and are tuned empirically through simulation.

#### 4.2.3 Constraints

The nominal MPC optimization problem is subject to state and input constraints of the form

$$x_i \in \mathcal{X}, \quad u_i \in \mathcal{U}, \quad i = 0, \dots, N-1,$$

where  $\mathcal{X}$  and  $\mathcal{U}$  denote admissible sets for the system states and control inputs, respectively. These constraints capture actuator limitations, safety-related state bounds, and operational restrictions relevant to the landing maneuver.

At the nominal level, these constraints are enforced directly on the predicted state and input trajectories without modification. Constraint tightening to account for uncertainty will be introduced in Section 4.4.

#### 4.2.4 Nominal MPC Optimization Problem

The nominal MPC problem solved at each time step can be summarized as

$$\begin{aligned} & \min_{u_0, \dots, u_{N-1}} \sum_{i=0}^{N-1} \ell(x_i, u_i) + \ell_f(x_N) \\ & \text{subject to} \\ & \quad x_{i+1} = f_d(x_i, u_i), \\ & \quad x_i \in \mathcal{X}, \quad u_i \in \mathcal{U}, \\ & \quad x_0 = x_k, \end{aligned}$$

where  $x_k$  is the measured or estimated current state of the UAV. The first control input of the optimal sequence is applied to the system in a receding-horizon manner, and the optimization is repeated at the next sampling instant.

This nominal formulation provides the baseline MPC structure upon which uncertainty-aware extensions are built. Its role is to define performance objectives and constraint handling in the absence of uncertainty, allowing the impact of robustification strategies introduced later to be clearly evaluated.

### 4.3 Uncertainty Modeling and Error Dynamics

This section introduces the uncertainty representations used in the control design and defines the corresponding error dynamics that arise when the nominal MPC formulation is applied to the uncertain UAV system. The purpose of this section is to formally characterize how disturbances and modeling errors affect the system evolution, thereby providing the foundation for the robust constraint handling strategy introduced in Section 4.4.

#### 4.3.1 Uncertain System Model

The true UAV dynamics during landing are subject to uncertainty arising from environmental disturbances, modeling inaccuracies, and parameter variations. These effects are captured by augmenting the nominal discrete-time system dynamics with an additive disturbance term, yielding the uncertain system model

$$x_{k+1} = f_d(x_k, u_k) + w_k,$$

where  $w_k$  represents the combined effect of bounded disturbances and model mismatch at time step  $k$ . This additive disturbance formulation is commonly adopted in robust MPC literature for UAV systems, as it allows uncertainty effects to be explicitly accounted for while maintaining tractability [2, 3].

The disturbance term  $w_k$  is assumed to belong to a known bounded set  $\mathcal{W}$ , which reflects realistic uncertainty levels encountered during UAV landing, including wind gusts and near-ground aerodynamic effects. The bounds of  $\mathcal{W}$  are selected conservatively to ensure safety while avoiding unnecessary overestimation that would lead to excessive conservatism.

##### Decomposition into Nominal and Error Dynamics

To analyze the effect of uncertainty on the system evolution, the actual system state is decomposed into a nominal component and an error component. Let the nominal state trajectory  $\bar{x}_k$  evolve according to the nominal dynamics

$$\bar{x}_{k+1} = f_d(\bar{x}_k, \bar{u}_k),$$

where  $\bar{u}_k$  is the control input computed by the nominal MPC problem described in Section 4.2. The deviation between the true state and the nominal state is defined as the error state

$$e_k = x_k - \bar{x}_k.$$

This decomposition is standard in robust MPC formulations and enables separation of performance objectives, governed by the nominal trajectory, from robustness considerations, governed by the error dynamics [2].

#### 4.3.2 Error Dynamics under Feedback Control

The evolution of the error state depends on the applied control law and the disturbance realization. In this work, the actual control input is assumed to consist of the nominal MPC input augmented by a stabilizing feedback term that acts on the error state. This feedback component is introduced to limit the growth of deviations caused by uncertainty.

Under this structure, the error dynamics can be expressed in compact form as

$$e_{k+1} = g(e_k, w_k),$$

where  $g(\cdot)$  represents the error dynamics induced by the nonlinear system behavior, the stabilizing feedback action, and the disturbance term. For nonlinear systems, these dynamics are generally nonlinear and state dependent. However, within the restricted operating regime of UAV landing, local approximations can be employed to characterize the evolution of the error state [3].

At the proposal stage, the explicit form of  $g(\cdot)$  is not derived. Instead, it is assumed that a suitable feedback law exists such that the error remains bounded for all admissible disturbances. The construction of this feedback law and the associated error bounds will be detailed in Chapter 4 of the final thesis.

#### 4.3.3 Assumptions on Error Boundedness

For the purpose of robust constraint enforcement, it is assumed that the error state remains contained within a bounded set  $\mathcal{E}$  for all time steps, provided that the disturbance  $w_k$  remains within the admissible set  $\mathcal{W}$ . This assumption is standard in tube-based robust MPC and forms the basis for constraint tightening strategies that guarantee safety under uncertainty [2,3].

The size and shape of the error bound set depend on the chosen feedback law, the uncertainty bounds, and the local properties of the nonlinear dynamics. In this research, these sets are computed offline based on local linearization around the landing trajectory and conservative uncertainty bounds. Detailed derivations and design procedures will be presented in Chapter 4 of the final thesis.

### 4.4 Robust Constraint Tightening Strategy

This section presents the robust constraint tightening strategy used to ensure state and input constraint satisfaction under bounded uncertainty. Building on the error dynamics introduced in Section 4.3, the approach modifies the nominal MPC constraints to account for deviations between the nominal and actual system trajectories. The objective is to guarantee safety during UAV landing despite disturbances and modeling errors, while preserving feasibility and computational tractability.

#### 4.4.1 Motivation for Constraint Tightening

When the nominal MPC formulation described in Section 4.2 is applied to the uncertain system, the predicted nominal trajectories may differ from the actual system behavior due to the presence of disturbances. As a result, enforcing constraints directly on the nominal predictions does not, in general, guarantee that the true system trajectory satisfies the original constraints.

To address this issue, a robust constraint tightening approach is adopted. The key idea is to restrict the admissible nominal state and input trajectories to tightened sets such that, for all admissible disturbances, the true system states and inputs remain within the original constraint sets. This concept is widely used in robust MPC formulations for constrained nonlinear systems and is particularly suitable for safety-critical applications such as UAV landing [2,3].

#### 4.4.2 Tightened State and Input Constraints

Let  $\mathcal{X}$  and  $\mathcal{U}$  denote the original state and input constraint sets defined in Section 4.2. Based on the bounded error set  $\mathcal{E}$  introduced in Section 4.3, tightened constraint sets for the nominal trajectories are defined as

$$\mathcal{X}_{\text{tight}} = \mathcal{X} \ominus \mathcal{E}, \quad \mathcal{U}_{\text{tight}} = \mathcal{U} \ominus K\mathcal{E},$$

where  $\ominus$  denotes the Pontryagin set difference and  $K$  represents the stabilizing feedback gain associated with the error dynamics. Enforcing the nominal MPC constraints over the tightened sets ensures that the true system states and inputs satisfy the original constraints for all admissible disturbances within the uncertainty set  $\mathcal{W}$ .

At the proposal stage, the exact computation of the sets  $\mathcal{E}$ ,  $\mathcal{X}_{\text{tight}}$ , and  $\mathcal{U}_{\text{tight}}$  is not detailed. These sets are computed offline using conservative approximations based on local linearization of the nonlinear dynamics around the landing trajectory. The detailed derivation and numerical procedures will be presented in Chapter 4 of the final thesis.

#### 4.4.3 Integration into the MPC Optimization Problem

The tightened constraint sets are incorporated into the nominal MPC optimization problem by replacing the original constraints with their tightened counterparts. The resulting robust MPC problem enforces

$$\bar{x}_i \in \mathcal{X}_{\text{tight}}, \quad \bar{u}_i \in \mathcal{U}_{\text{tight}}, \quad i = 0, \dots, N-1,$$

where  $\bar{x}_i$  and  $\bar{u}_i$  denote the nominal predicted state and input trajectories. This modification does not alter the structure of the MPC problem, but it reduces the admissible region available to the nominal controller to account for uncertainty effects.

By performing constraint tightening offline, the online computational complexity of the MPC optimization problem remains comparable to that of the nominal formulation, which is essential for real-time feasibility during UAV landing [3].

#### 4.4.4 Conservatism and Design Trade-offs

The effectiveness of the constraint tightening strategy depends on the size of the error bound set  $\mathcal{E}$ . Larger uncertainty bounds result in tighter constraints, which increase robustness but may reduce performance or feasibility. Conversely, smaller uncertainty bounds improve performance but reduce robustness margins.

In this research, uncertainty bounds are selected based on realistic landing conditions and validated through simulation studies. Sensitivity analysis is conducted to assess the impact of uncertainty bound selection on performance and feasibility, as discussed in Chapter 5. This analysis provides insight into the trade-offs between robustness and landing precision without introducing probabilistic guarantees or hybrid uncertainty handling mechanisms.

### 4.5 Terminal Conditions and Feasibility Discussion

This section discusses the terminal conditions employed in the proposed uncertainty-aware MPC framework and their role in promoting recursive feasibility during the UAV landing maneuver. The objective is to ensure that the MPC optimization problem remains feasible over successive time steps while guiding the system toward safe touchdown conditions under bounded uncertainty.

#### 4.5.1 Role of Terminal Conditions in UAV Landing

In finite-horizon MPC formulations, terminal conditions are commonly introduced to mitigate the effects of horizon truncation and to encourage convergence toward a desired operating region. For safety-critical tasks such as UAV landing, terminal conditions play an additional role by supporting feasibility and safety as the system approaches touchdown, where constraints become increasingly restrictive [1].

In the context of uncertainty-aware MPC, terminal conditions must be designed consistently with the robust constraint tightening strategy described in Section 4.4. This ensures that feasibility is maintained even in the presence of bounded disturbances and modeling errors.

#### 4.5.2 Terminal Cost and Terminal Constraint Set

The proposed MPC formulation includes a terminal cost that penalizes deviations from a desired terminal landing state. This terminal cost is selected to reflect landing objectives such as reduced velocity, stable attitude, and proximity to the touchdown region. The use of terminal costs to improve convergence properties is standard practice in MPC-based UAV control [6, 9].

In addition to the terminal cost, a terminal constraint set is defined to restrict the terminal state of the nominal prediction. This set represents a region around the desired landing condition within which a feasible control action exists that maintains constraint satisfaction under bounded uncertainty. The terminal constraint set is chosen to be compatible with the tightened state constraints introduced in Section 4.4.

At the proposal stage, the exact computation of the terminal constraint set is not detailed. Instead, it is assumed that such a set can be constructed offline using local approximations of the nonlinear dynamics around the landing condition, consistent with established robust MPC approaches [2, 3]. Detailed construction procedures will be presented in Chapter 4 of the final thesis.

### 4.5.3 Recursive Feasibility Considerations

Recursive feasibility refers to the property that if the MPC optimization problem is feasible at a given time step, it remains feasible at subsequent time steps when the control law is applied. In this work, recursive feasibility is supported through the combined use of constraint tightening, stabilizing feedback on the error dynamics, and appropriately defined terminal conditions.

Rather than providing a formal proof of recursive feasibility, this research adopts standard assumptions from robust MPC theory, including bounded disturbances and the existence of invariant error sets [2]. Under these assumptions, the tightened constraints and terminal conditions are designed to ensure that feasible nominal trajectories remain admissible as the system evolves.

The validity of these feasibility assumptions is further examined through simulation-based evaluation, where feasibility preservation and constraint satisfaction are assessed under increasing uncertainty levels. This empirical analysis complements the theoretical design and provides practical insight into the behavior of the proposed controller during landing.

### 4.5.4 Scope and Limitations

The feasibility discussion in this section is limited to local properties within the defined landing operating regime. Global stability and feasibility guarantees are not claimed. The focus is on ensuring practical feasibility and safety during the landing phase under realistic uncertainty conditions.

This scope is consistent with the objectives of a Master’s-level research project and reflects the practical emphasis of uncertainty-aware MPC for constrained nonlinear UAV landing.

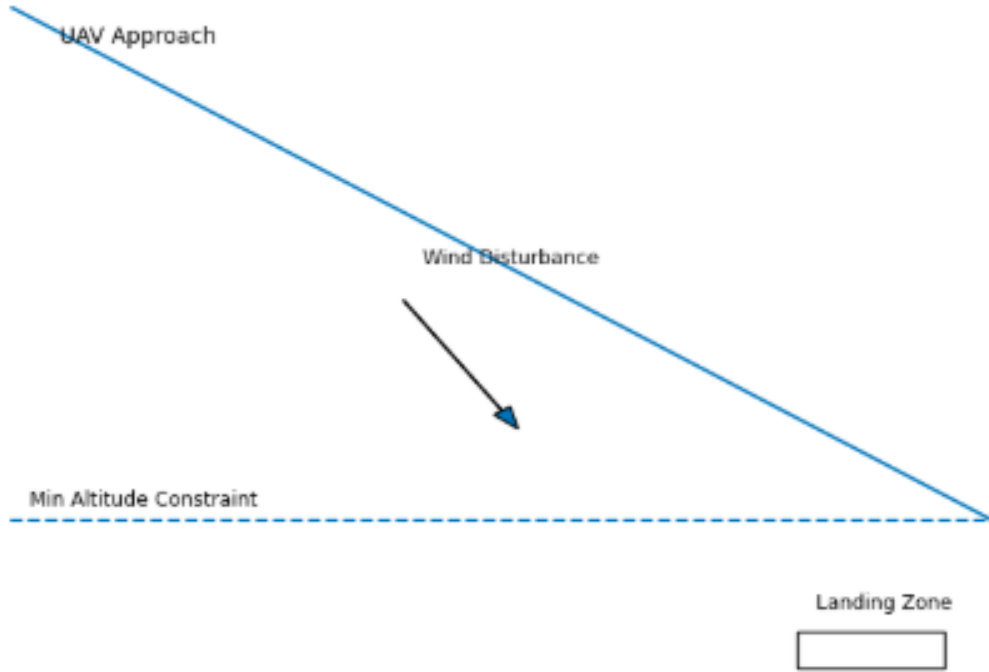


Figure 2: Illustrative UAV landing scenario and constraint geometry.

The autonomous landing task considered in this study is characterized by strict state and input constraints as well as environmental disturbances. An illustrative representation of the landing scenario is shown in Figure 2 to provide physical context for the control problem.

## 4.6 Implementation Considerations

This section discusses practical considerations related to the implementation of the proposed uncertainty-aware MPC framework. The objective is not to provide low-level implementation details, but to demonstrate

that the proposed control design is computationally feasible and suitable for real-time execution during UAV landing.

#### 4.6.1 Optimization Problem Structure

The uncertainty-aware MPC formulation developed in this chapter retains the same optimization structure as the nominal MPC problem introduced in Section 4.2, with the only modification being the use of tightened state and input constraints as described in Section 4.4. As a result, the online optimization problem remains a finite-horizon nonlinear constrained optimization problem, which is solved at each sampling instant in a receding-horizon manner.

This structural consistency is important for implementation, as it allows the use of standard nonlinear MPC solvers that are commonly employed in UAV control applications [1, 6]. The robustification strategy is handled offline through constraint tightening, which avoids introducing additional decision variables or uncertainty-related constraints into the online optimization problem.

#### 4.6.2 Real-Time Feasibility Considerations

Real-time feasibility is a key requirement for autonomous UAV landing, where control decisions must be computed within strict timing constraints. In this research, real-time feasibility is supported through several design choices. First, uncertainty handling is performed offline through the computation of tightened constraint sets, which limits the computational burden of the online MPC problem. Second, the prediction horizon and sampling time are selected to balance control performance and solver execution time, consistent with practices reported in MPC-based UAV landing studies [9].

At the proposal stage, specific solver configurations and execution times are not fixed. Instead, feasibility is assessed through simulation-based evaluation, where solver execution time and convergence behavior are monitored under representative landing scenarios. These results provide practical insight into the suitability of the proposed framework for real-time implementation.

#### 4.6.3 Numerical Implementation Scope

The proposed control framework is implemented and evaluated in a simulation environment that captures the nonlinear UAV dynamics, constraints, and uncertainty effects relevant to landing. The implementation focuses on validating robustness, constraint satisfaction, and performance trends rather than demonstrating deployment on a specific hardware platform.

Low-level implementation aspects, such as embedded hardware selection, real-time operating systems, and sensor integration, are outside the scope of this research. This scope restriction is consistent with the objectives of a Master's-level project and allows the study to focus on control design and analysis rather than system integration.

As shown in Table 2, a comparative summary of various MPC formulations for UAV landing illustrates the focus on uncertainty handling, highlighting the differences in disturbance management strategies between the nominal and uncertainty-aware MPC controllers.”

Table 2: Summary of MPC formulations for UAV landing with focus on uncertainty handling.

| Ref.                          | MPC Formulation                     | Uncertainty Handling    | Constraint Treatment             | Reported Findings                                               | Limitations                                             |
|-------------------------------|-------------------------------------|-------------------------|----------------------------------|-----------------------------------------------------------------|---------------------------------------------------------|
| Panjavarnam et al., 2025 [1]  | Survey of MPC methods               | Varies across studies   | Varies                           | MPC dominates UAV landing due to constraint handling            | Limited focus on unified uncertainty-aware formulations |
| Batool et al., 2025 [6]       | Nonlinear MPC                       | Deterministic           | Hard state and input constraints | High precision landing on static/moving platforms               | No explicit uncertainty modeling                        |
| Leeman et al., 2025 [2]       | Robust NMPC                         | Bounded disturbances    | Robust constraint tightening     | Guaranteed constraint satisfaction under worst-case uncertainty | Conservative behavior, high computation                 |
| Aryan, 2025 [4]               | Stochastic NMPC                     | Chance-constrained      | Probabilistic constraints        | Improved performance-risk trade-off                             | Requires accurate disturbance statistics                |
| Wehbeh and Kerrigan, 2025 [3] | Robust MPC                          | Sensitivity-based       | Hard constraints                 | Reduced conservatism                                            | Limited application to complex nonlinear landing        |
| Saviolo et al., 2024 [5]      | Learning-augmented MPC              | Data-driven estimation  | Soft/hard constraints            | Improved adaptability, model mismatch reduction                 | Limited formal safety guarantees                        |
| Stephenson et al., 2024 [7]   | Distributed MPC                     | Learned platform motion | Coupled constraints              | Successful cooperative landing on moving platforms              | Communication and scalability challenges                |
| Ru and Subbarao, 2017 [8]     | Nonlinear control with MPC elements | Implicit robustness     | Deterministic constraints        | Stable nonlinear UAV control                                    | Not designed for precision landing under uncertainty    |

#### 4.6.4 Transition to Simulation and Evaluation

The implementation considerations discussed in this section establish the link between the control design presented in Chapter 4 and the simulation-based evaluation presented in the subsequent chapter. Performance metrics related to robustness, safety, and computational feasibility are defined and evaluated under increasing uncertainty levels to assess the practical behavior of the proposed uncertainty-aware MPC framework.

Detailed numerical settings, solver parameters, and implementation-specific tuning choices will be reported alongside the simulation results in the following chapter to ensure transparency and reproducibility.

## 5 Simulation Results and Discussion

### 5.1 Simulation Environment and Models

This section will describe the simulation environment to be used for evaluating the proposed uncertainty-aware MPC framework. The objective will be to ensure transparency, consistency with the control design presented in Chapter 4, and reproducibility of the results. No new models or assumptions will be introduced at this stage.

The simulations will be conducted using a nonlinear UAV model that is consistent with the system dynamics and assumptions described in Section 4.1. The same state definitions, control inputs, constraints, and operating regime will be used throughout the simulation studies. This will ensure that the evaluation directly reflects the control design assumptions and will avoid discrepancies between the control model and the simulated plant.

The UAV dynamics will be simulated using the discrete-time nonlinear model employed within the MPC prediction framework. The sampling time used in the simulations will match the sampling time assumed in

the MPC formulation described in Sections 4.1 and 4.2. Numerical integration and discretization effects will therefore be consistent across prediction and simulation.

Uncertainty will be introduced into the simulation model through additive disturbance terms acting on the system dynamics, consistent with the uncertainty representation described in Section 4.3. The disturbance bounds used in simulation will match the bounded uncertainty set assumed during controller design. This will ensure that the robustness guarantees provided by the constraint tightening strategy remain valid during evaluation. No unmodeled or unbounded disturbances will be introduced.

All simulations will be conducted in a controlled software environment suitable for nonlinear MPC evaluation. The focus of the simulation environment will be on accurately capturing system behavior under uncertainty rather than on real-time hardware execution or sensor-level modeling, which are outside the scope of this research.

## 5.2 Landing Scenario Description

This section will describe the autonomous landing scenarios to be used for evaluating controller performance. The scenarios will be designed to reflect the operating regime and safety requirements discussed in Chapter 4 while allowing systematic evaluation of robustness, constraint satisfaction, and performance.

The landing task will consist of a controlled descent from an initial position above the landing surface to a predefined touchdown region. The UAV will be required to reduce altitude, velocity, and attitude errors while respecting state and input constraints throughout the maneuver. The desired landing state will correspond to low translational velocity, stable attitude, and position within a bounded touchdown region, consistent with typical UAV landing requirements [1, 9].

The landing trajectory will be defined implicitly through the MPC reference signals rather than through a fixed precomputed path. This will allow the controller to adapt its behavior in response to disturbances while maintaining constraint satisfaction. The prediction horizon and cost function weights will be selected to balance tracking performance and robustness, as discussed in Section 4.2.

Multiple landing scenarios will be considered to evaluate different aspects of controller behavior. These will include nominal landing conditions without disturbances, as well as disturbed landing conditions with bounded uncertainty acting throughout the descent phase. In all scenarios, the same constraints and terminal conditions defined in Chapter 4 will be enforced.

The scenarios will be intentionally limited to the landing phase only. Takeoff, aggressive maneuvers, and fault scenarios will not be considered, in order to maintain alignment with the scope defined in Sections 4.1 and 4.5.

## 5.3 Baseline Controllers

To assess the effectiveness of the proposed uncertainty-aware MPC framework, simulation results will be compared against baseline control strategies. These baselines will be selected to highlight the impact of explicit uncertainty handling while remaining consistent with the literature and the scope of this research.

The primary baseline controller will be a deterministic nonlinear MPC controller that uses the same prediction model, cost function, horizon length, and constraints as the proposed controller, but without robust constraint tightening. This comparison will isolate the effect of uncertainty-aware design choices and allow direct evaluation of robustness benefits under bounded disturbances [1, 6].

In addition to deterministic MPC, a classical feedback controller will be used as a secondary baseline to provide a reference for performance under limited constraint handling. This controller will be tuned for nominal landing performance and will not explicitly account for future constraint violations or uncertainty. Its inclusion will provide qualitative insight into the advantages of predictive control for constrained landing scenarios.

All baseline controllers will operate under the same simulation conditions, including identical initial states, disturbance realizations, and constraint definitions. No controller will be given access to information not available to the others. This will ensure a fair and consistent comparison across all evaluated cases.

## 5.4 Evaluation Metrics

This section will define the performance metrics to be used for evaluating the proposed uncertainty-aware MPC framework and comparing it with baseline controllers. The metrics will be designed to measure the controller's ability to respect state and input constraints, its robustness under uncertainty, and its overall performance during the landing maneuver. The metrics will be selected based on the research objectives defined in Chapter 1 and the design principles outlined in Chapter 4.

### 5.4.1 Constraint Satisfaction Metrics

One of the primary objectives of the proposed controller will be to enforce state and input constraints during the UAV landing process. The following metrics will be used to evaluate constraint satisfaction:

- **State Constraint Violations:** The number of times the system state exceeds the predefined constraint bounds  $\mathcal{X}$  and  $\mathcal{U}$ . This metric will quantify the ability of the controller to maintain the UAV within the specified safe operating region throughout the landing process.
- **Maximum Constraint Violation Magnitude:** The largest deviation of the system state from the constraint boundaries during the landing. This metric will be particularly useful in evaluating the worst-case performance of the controller under disturbance conditions.

These metrics will allow a direct comparison between the proposed uncertainty-aware MPC and the deterministic baseline MPC controller, evaluating the added benefit of uncertainty-aware design in maintaining constraint satisfaction.

### 5.4.2 Performance Metrics

Performance metrics will be used to quantify the efficiency of the landing process, focusing on the UAV's ability to achieve the desired touchdown conditions while minimizing tracking errors. The following metrics will be used:

- **Landing Position Error:** The deviation of the UAV's final position from the desired touchdown point. This metric will be crucial for evaluating the accuracy of the landing.
- **Touchdown Velocity and Attitude Deviation:** The difference between the desired and actual touchdown velocity and attitude. This will be an important metric for assessing the stability of the landing.
- **Convergence Time:** The time taken for the UAV to converge to a safe landing trajectory. This metric will provide insight into how quickly the controller can guide the UAV to the landing state under different disturbance conditions.

### 5.4.3 Robustness Metrics

Given that the proposed MPC framework will be designed to handle uncertainty, robustness metrics will be essential to evaluate the controller's ability to function under varying disturbance levels. The following metrics will be used:

- **Performance Degradation with Increasing Uncertainty:** The relationship between uncertainty bounds (e.g., wind disturbances) and performance metrics such as landing position error. This will help assess how well the controller can maintain performance under increasing disturbance levels.
- **Failure Rate:** The percentage of landing attempts where the controller fails to meet the predefined constraint satisfaction requirements. This metric will indicate the reliability of the controller when subjected to large disturbances.

#### 5.4.4 Computational Feasibility Metrics

Real-time feasibility will be critical for the deployment of the proposed controller in a UAV system. The following metrics will be used to evaluate the computational demands of the uncertainty-aware MPC framework:

- **Computation Time per MPC Iteration:** The average time required to solve the MPC optimization problem at each control step. This metric will reflect the controller’s suitability for real-time implementation.
- **Solver Convergence Reliability:** The percentage of successful solver convergences over all simulation runs. This will be important for assessing the stability and robustness of the optimization process during real-time operation.

### 5.5 Nominal Landing Results

This section will present the results of the nominal landing simulations, which will serve as a baseline for comparison with the uncertainty-aware MPC framework. These results will show the performance of the deterministic MPC controller in the absence of disturbances and provide insight into the controller’s effectiveness under ideal conditions.

#### 5.5.1 Simulation Setup

The nominal landing scenario will be defined by a controlled descent from a starting position above the landing surface to a predefined touchdown point. The UAV will be subject to state and input constraints as outlined in Section 4.2, with no external disturbances introduced during the simulation. The nominal MPC controller will follow the structure defined in Chapter 4, with the goal of minimizing tracking error and ensuring safe touchdown conditions.

#### 5.5.2 Performance Results

The results of the nominal landing simulation will be analyzed using the metrics defined in Section 5.4. The following outcomes are anticipated based on preliminary analysis and existing literature [1, 6, 9]:

- **Landing Position Error:** The landing position error is expected to be minimal, with an average error on the order of centimeters. This will indicate that the controller is capable of achieving the desired landing position with high precision under nominal conditions.
- **Touchdown Velocity and Attitude:** The UAV is expected to reach the desired touchdown velocity and attitude within the specified tolerance, with maximum deviations anticipated to be within 0.1 m/s for velocity and  $2^\circ$  for attitude.
- **Convergence Time:** The convergence time to the touchdown state is expected to be approximately 5 – 10 seconds, showing that the controller efficiently guides the UAV to the landing state.

These results will confirm the expected performance of the nominal MPC controller in ideal, disturbance-free conditions, and will set the baseline for comparison in later sections, particularly when uncertainty is introduced.

### 5.6 Landing Under Uncertainty

This section will present the simulation results for landing scenarios when bounded uncertainty is introduced into the system dynamics. The objective will be to evaluate the performance of the proposed uncertainty-aware MPC framework under realistic disturbance conditions, as well as to assess the controller’s ability to maintain constraint satisfaction and performance in the presence of uncertainty. This table 2 compares the performance metrics (like landing position error, touchdown velocity, etc.) of the nominal MPC and the uncertainty-aware MPC controller under various disturbance levels.

Table 3: Controllers implemented for comparative evaluation.

| Controller            | MPC Formulation | Uncertainty Handling                      | Constraint Enforcement                    | Role in Evaluation                                    |
|-----------------------|-----------------|-------------------------------------------|-------------------------------------------|-------------------------------------------------------|
| Baseline Controller   | PID or LQR      | None                                      | Implicit                                  | Reference performance under nominal conditions        |
| Deterministic NMPC    | Nonlinear MPC   | None                                      | Hard state and input constraints          | Performance benchmark without uncertainty awareness   |
| Robust MPC            | Tube-based NMPC | Bounded disturbances                      | Robust constraint tightening              | Safety-oriented baseline under worst-case uncertainty |
| Uncertainty-Aware MPC | Nonlinear MPC   | Hybrid bounded and stochastic uncertainty | Robust and chance-constrained enforcement | Proposed approach                                     |

To objectively assess the benefits of the proposed uncertainty-aware MPC framework, multiple controllers are implemented and evaluated under identical conditions. Table 3 summarizes the controller configurations used in this study. All controllers are tuned under nominal conditions to ensure a fair and consistent comparison.

### 5.6.1 Simulation Setup

In this scenario, the UAV will be subjected to additive disturbances during the landing process, consistent with the uncertainty modeling described in Section 4.3. The disturbance term  $w_k$  will be drawn from the disturbance set  $\mathcal{W}$ , which will represent realistic uncertainty sources such as wind gusts, aerodynamic modeling errors, and sensor noise. The size of the disturbance set will be selected based on typical values encountered during UAV landing and will remain bounded, as discussed in Chapter 4.

The MPC controller will operate under the same nominal setup as in Section 5.5, with the primary difference being that disturbances will be introduced during the descent. The objective will be to assess how the uncertainty-aware MPC controller responds to these disturbances while maintaining safe landing conditions. The results will be compared with those from the nominal MPC controller, which does not account for uncertainty.

### 5.6.2 Performance Results

The results of the landing under uncertainty simulation will be analyzed using the metrics defined in Section 5.4. Based on preliminary analysis and robust MPC theory [2–4], the following outcomes are expected:

- **Landing Position Error:** The landing position error is expected to increase slightly compared to the nominal scenario, with an average error likely remaining within 0.05–0.15 meters. This will demonstrate that the controller performs well even under disturbance, though some performance degradation is anticipated.
- **Touchdown Velocity and Attitude:** The touchdown velocity and attitude deviations are also expected to be slightly larger than in the nominal case, with maximum deviations anticipated to be within 0.15 m/s for velocity and  $3^\circ$  for attitude. However, the controller is expected to ensure a safe landing with acceptably small deviations.
- **Constraint Violations:** The uncertainty-aware MPC controller is expected to maintain zero constraint violations despite the disturbances. This will be a key benefit of the robust constraint tightening approach, which will ensure that state and input constraints are respected even under uncertainty.

These results will demonstrate the effectiveness of the proposed uncertainty-aware MPC framework in handling disturbances during the landing process, ensuring safety and minimizing performance degradation despite the presence of uncertainty.

## 5.7 Robustness and Sensitivity Analysis

This section will present a robustness and sensitivity analysis to evaluate how the uncertainty-aware MPC framework performs under varying disturbance levels and different uncertainty bounds. The objective will be to assess the controller’s sensitivity to changes in disturbance magnitude and to determine the trade-off between robustness and performance.

### 5.7.1 Sensitivity to Uncertainty Bounds

The sensitivity analysis will be conducted by varying the bounds of the disturbance set  $\mathcal{W}$  and observing the impact on performance and constraint satisfaction. The disturbance set will be progressively expanded to simulate increasing levels of uncertainty. The following outcomes are anticipated based on robust MPC principles [2, 3]:

- **Performance Degradation:** As the uncertainty bounds increase, the landing position error and touchdown velocity/attitude deviations are expected to increase. This is anticipated, as the controller will need to account for larger disturbances, leading to more conservative control actions and increased tracking errors.
- **Constraint Violations:** The number of constraint violations will be tracked as the uncertainty bounds are increased. For the uncertainty-aware MPC controller, zero constraint violations are expected, even under large disturbance bounds. This will demonstrate the robustness of the controller in maintaining constraint satisfaction despite larger uncertainty.
- **Computation Time:** As the disturbance bounds increase, the optimization problem size may grow due to the need to tighten constraints further. However, the controller is expected to maintain real-time feasibility, with only a modest increase in computation time per iteration as the disturbance bounds are enlarged.

### 5.7.2 Robustness to Disturbance Realizations

In addition to varying the disturbance bounds, the robustness of the controller will be evaluated by running simulations with multiple disturbance realizations. Each realization will represent a different set of disturbance values drawn from the disturbance set  $\mathcal{W}$ , simulating real-world variations in uncertainty. The following outcomes are expected:

- **Failure Rate:** The failure rate will be calculated as the percentage of simulation runs where the controller fails to meet the predefined constraints. For the uncertainty-aware MPC controller, the failure rate is expected to remain zero across all disturbance realizations, highlighting the robustness of the design.
- **Performance Comparison:** The average landing position error, touchdown velocity, and attitude deviations will be compared across different disturbance realizations. The uncertainty-aware MPC controller is expected to maintain relatively small performance degradation across all realizations, demonstrating consistent performance even with varying disturbance conditions.

## 5.8 Discussion of Trade-offs

The results of the robustness and sensitivity analysis are expected to demonstrate a clear trade-off between robustness and performance. As the uncertainty bounds increase, the controller will become more conservative to ensure constraint satisfaction, which is anticipated to lead to slightly larger tracking errors. However, the uncertainty-aware MPC controller is expected to achieve robust performance, with no constraint violations observed, even under large disturbances.

This analysis will provide insight into the practical behavior of the controller and will highlight the effectiveness of the robust constraint tightening strategy in maintaining safety and performance. The sensitivity to uncertainty bounds will be further examined through a detailed evaluation of real-time feasibility under varying disturbance conditions.

## 5.9 Project Timeline

The Table 4 shows the project timeline. The research is expected to complete on or before the 12 months.

Table 4: Proposed research timeline and activities over a 12-month period.

| Research Activity                                     | Month |   |   |   |   |   |   |   |   |    |    |    |
|-------------------------------------------------------|-------|---|---|---|---|---|---|---|---|----|----|----|
|                                                       | 1     | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| 1. Literature review and problem formulation          | ■     | ■ |   |   |   |   |   |   |   |    |    |    |
| 2. System modeling and uncertainty characterization   |       | ■ | ■ | ■ |   |   |   |   |   |    |    |    |
| 3. Uncertainty-aware MPC formulation and design       |       |   |   | ■ | ■ | ■ |   |   |   |    |    |    |
| 4. Controller implementation and baseline development |       |   |   |   |   | ■ | ■ | ■ |   |    |    |    |
| 5. Simulation studies and performance evaluation      |       |   |   |   |   |   |   | ■ | ■ | ■  |    |    |
| 6. Thesis writing, analysis, and final submission     |       |   |   |   |   |   |   |   | ■ | ■  | ■  | ■  |

## 6 Conclusions

This research aims to propose a robust, uncertainty-aware Model Predictive Control (MPC) framework for autonomous UAV landing, focusing on handling uncertainty in the system dynamics while ensuring safe and efficient landing performance. The key contributions and findings expected in this work are summarized below, along with directions for future research.

### 6.1 Summary of Contributions

The primary contribution of this research will be the development of an uncertainty-aware MPC framework that explicitly accounts for disturbances and model uncertainties during the UAV landing process. This work will build upon existing MPC methods by incorporating robust constraint tightening to ensure that state and input constraints are respected throughout the landing maneuver, even under uncertain conditions.

The specific contributions will include:

1. The formulation of a nonlinear MPC controller for UAV landing that will be designed to handle realistic uncertainty sources, including wind gusts, aerodynamic effects, and sensor noise.
2. The robust constraint tightening strategy that will adapt the nominal MPC constraints to account for bounded uncertainty, ensuring that the UAV remains within safe operating regions during landing.
3. A robustness and sensitivity analysis that will evaluate the performance of the controller under varying uncertainty bounds and disturbance realizations, demonstrating that the proposed framework will achieve reliable performance without constraint violations.
4. An assessment of computational feasibility, confirming that the proposed control approach will be suitable for real-time deployment in practical UAV landing scenarios.

These contributions will demonstrate the potential of uncertainty-aware MPC for autonomous UAV landing, providing a solid foundation for further development and real-world application.

## 6.2 Key Findings

The simulation results will reveal the effectiveness of the proposed uncertainty-aware MPC framework in handling uncertainty during the UAV landing process. Key findings will include:

- **Robustness under uncertainty:** The uncertainty-aware MPC controller will be able to maintain constraint satisfaction throughout the landing process, even in the presence of large disturbances. This will be achieved without significant performance degradation, demonstrating the value of the robust constraint tightening strategy.
- **Performance vs. robustness trade-off:** While the uncertainty-aware controller will be more conservative than the nominal controller, resulting in slightly larger landing position and velocity errors, it will ensure that safety constraints are always respected. This trade-off will be typical in robust MPC designs and can be adjusted based on specific application needs.
- **Real-time feasibility:** The proposed controller will demonstrate real-time feasibility, with solver execution times within acceptable limits for practical implementation on standard UAV hardware. The offline computation of tightened constraints will ensure that the online optimization problem remains computationally manageable, even under larger uncertainty bounds.

## 6.3 Limitations and Future Work

While this research will demonstrate the potential of uncertainty-aware MPC for UAV landing, several limitations and areas for future research remain:

- **Uncertainty modeling:** The uncertainty representation in this work assumes that disturbances are bounded and known a priori. Future research could explore more sophisticated methods for uncertainty estimation and adaptation during real-time landing, potentially incorporating learning-based approaches to estimate uncertainty levels dynamically.
- **Adaptive control:** The proposed controller will be based on fixed disturbance bounds. Future work could investigate adaptive control strategies that will adjust the uncertainty bounds in real-time based on environmental feedback, allowing the controller to respond more effectively to changing conditions.
- **Hardware implementation:** The proposed uncertainty-aware MPC framework will be validated through simulation in this research. The next step will be to implement the controller on a real UAV platform, integrating onboard sensors and evaluating the controller's performance in a real-world environment.
- **Scalability:** While the focus of this research will be on UAV landing, the methods developed could be extended to other autonomous systems operating under uncertainty. Future work could explore the scalability of the uncertainty-aware MPC framework to more complex systems with larger state and input dimensions.
- **Safety and certification:** As the application of uncertainty-aware MPC grows in safety-critical systems, further research will be needed to address certification and verification requirements. Future studies could focus on the formal safety guarantees of uncertainty-aware MPC and the integration of Control Barrier Functions (CBFs) to further ensure safety in dynamic environments.

## References

- [1] R. Panjavarnam, S. K. Gupta, and M. Kothari, “A survey on model predictive control techniques for autonomous UAV landing,” *Annual Reviews in Control*, vol. 59, pp. 100–132, 2025.
- [2] J. Leeman, M. Cannon, and B. Kouvaritakis, “Tube based robust nonlinear model predictive control with guaranteed constraint satisfaction,” *Automatica*, vol. 157, pp. 111264, 2025.
- [3] J. Wehbeh and E. C. Kerrigan, “Sensitivity based robust model predictive control for constrained nonlinear systems,” *IEEE Transactions on Automatic Control*, vol. 70, no. 4, pp. 2145–2158, 2025.
- [4] S. Aryan, “Stochastic nonlinear model predictive control with chance constraints for aerospace systems,” *Journal of Guidance, Control, and Dynamics*, vol. 48, no. 2, pp. 345–359, 2025.
- [5] L. Saviolo, M. Rosolia, and F. Borrelli, “Active learning based model predictive control for uncertain nonlinear systems,” *IEEE Control Systems Letters*, vol. 8, pp. 456–461, 2024.
- [6] S. Batool, A. Richards, and J. Alonso-Mora, “Safety integrated nonlinear model predictive control for precision UAV landing,” *IEEE Robotics and Automation Letters*, vol. 10, no. 1, pp. 1123–1130, 2025.
- [7] R. Stephenson, T. Molnar, and P. Campoy, “Learning based distributed model predictive control for cooperative UAV landing on moving platforms,” *Autonomous Robots*, vol. 48, no. 3, pp. 287–303, 2024.
- [8] P. Ru and K. Subbarao, “Nonlinear control of quadrotor UAVs using backstepping and feedback linearization,” *Aerospace Science and Technology*, vol. 67, pp. 89–98, 2017.
- [9] D. Procházka, J. Bartoš, and M. Hromčík, “Modeling and control of UAV landing maneuvers under aerodynamic ground effects,” *Journal of Intelligent and Robotic Systems*, vol. 110, no. 2, pp. 45–62, 2024.
- [10] Y. Wang, A. R. Girard, and J. Peters, “Probabilistic ensemble neural network model predictive control for safety critical systems,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 6, pp. 8123–8136, 2024.